

Sydney Girls High School

2003

TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics

Extension 1

This is a trial paper ONLY.
It does not necessarily
reflect the format or the
contents of the 2003 HSC
Examination Paper in this
subject.

General Instructions

- ♦ Reading Time – 5 mins
- ♦ Working Time – 2 hours
- ♦ Attempt ALL questions
- ♦ ALL questions are of equal value
- ♦ All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- ♦ Standard integrals are supplied
- ♦ Board-approved calculators may be used.
- ♦ Diagrams are not to scale
- ♦ Each question attempted should be started on a new sheet. Write on one side of the paper only.

Marks

Question 1 (12 marks)

a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$ (1)

b) Evaluate i) $\int_0^{\pi} \sin^2 x dx$ (2)

ii) $\int_{-1}^0 x\sqrt{1+x} dx$ (3)

using the substitution $u = 1 + x$

c) Find the point which divides the line joining A (1, 3) and B (-2, 6) externally in the ratio 2 : 1 (2)

d) Write $3 \cos \theta + 4 \sin \theta$ in the form $R \cos (\theta - \alpha)$ and Hence solve $3 \cos \theta + 4 \sin \theta = 2$, $0 \leq \theta \leq 2\pi$ (4)
(Give you answer correct to 2 decimal places)

Marks

Question 2 (12 marks)

(a) Solve $\frac{-2x}{x-1} < 1$ (3)

(b) Given that $x = 1.7$ is a first approximation to the positive root of $x = 2 \sin x$, use Newton's method once to find a second approximation to this root. (Correct to 1 decimal place) (3)

(c) Find the size of the acute angle between the lines $4x - 3y + 1 = 0$ and $x + 4y + 1 = 0$. (Give answer to the nearest degree). (3)

(d) A polynomial is given by $P(x) = x^3 + ax^2 + bx - 18$ (3)
Find the values of a and b if $(x+2)$ is a factor of $P(x)$ and if -24 is the Remainder when $P(x)$ is divided by $(x-1)$.

Marks

Question 3 (12 marks)

- (a) For the function $y = 2\sin^{-1}x$, find the equation of the tangent to the curve at the point where $x = \frac{1}{\sqrt{2}}$. (3)

- (b) The acceleration of a particle moving in a straight line is given by (5)

$$\frac{d^2x}{dt^2} = 2x - 3$$

where x is the position (in metres) from the origin 0 and t is the time in seconds. Initially the particle is at rest at $x = 4$ m.

- If the velocity is v m/s show that $v^2 = 2x^2 - 6x - 8$
- Show that the particle does not pass through the origin.
- Find the position when $v = 10$ m/s

- (c) The Volume (V) of a sphere of radius r cm is increasing at a constant rate of 200 cm^3 per second. (4)

- Find $\frac{dr}{dt}$ in terms of r
- Hence find the rate of increase of the surface Area (A) when the radius of the sphere is 50 cm.

Question 4 (12 marks)

- (a) Prove the identity $\frac{2 \tan \Theta}{1 + \tan^2 \Theta} = \sin 2\Theta$ (1)

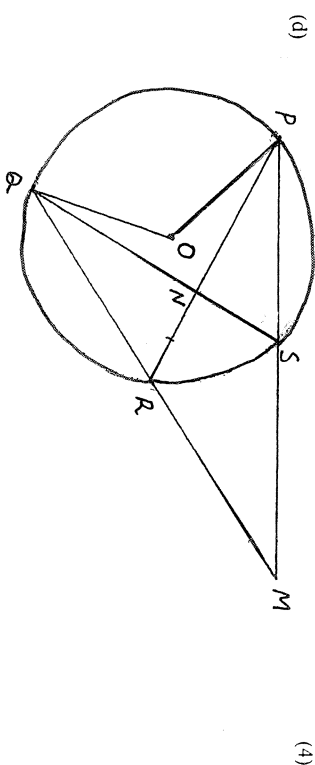
- (b) For the function $f(x) = 2 \cos^{-1} \frac{x}{3}$ (3)

- Evaluate $f(0)$
- State the domain and range of $y = f(x)$
- Sketch the graph of $y = f(x)$

- (c) A particle moves in simple harmonic motion about the origin 0. Its position x metres from 0 at time t seconds is given by, (4)

$$x = 3 \cos \left(2t + \frac{\pi}{3} \right)$$

- Find the acceleration in terms of position.
- Find its amplitude
- State the position x for maximum velocity.
- Find the maximum velocity.



0 is the centre of a circle and $\angle POQ = \Theta^\circ$. Lines PS and QR produced intersect at M and lines PR and QS intersect at N

- Copy this diagram into your exam booklet
- Prove that $\angle PRM = \left(180 - \frac{1}{2} \Theta \right)^\circ$
- Prove that $\angle PNQ + \angle PMQ = \Theta$

Question 5 (12 marks)

- (a) If α, β, γ are the roots of $x^3 - 3x + 1 = 0$ (3)

Find i $\alpha + \beta + \gamma$

ii $\alpha\beta\gamma$

iii $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

- (b) i. Factorise the polynomial $f(x) = 3x^3 - 7x^2 + 4$ (3)

ii. Hence solve $f(x) = 0$

- (c) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$ (3)

i. Derive the equation of the tangent at P

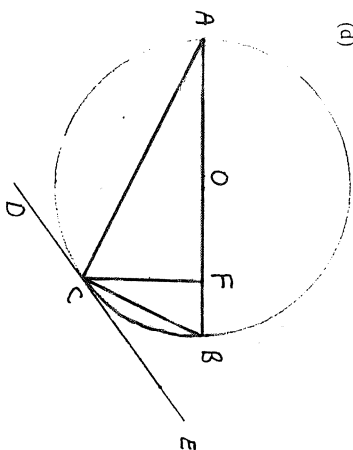
ii. Find the coordinates of the point of intersection T of the tangents to the parabola at P and Q

iii. If these tangents intersect at 45° show that $p = 1 + q + pq$, if $p > q$

- (d) CF is perpendicular to AB (3)

O is the centre of the circle

DCE is a tangent at C



- i. Copy this diagram into your workbook
ii. Prove that BC bisects angle FCE

Question 6 (12 marks)

- (a) In a population study, the population N is given by the equation (3)

$$N = 200 + Ae^{kt}$$

Initially $N = 300$ and when $t = 3$ seconds, $N = 500$

- i. Find the values of A and k (correct to 4 decimal places)
ii. Find the population after 5 seconds

- (b) i. If $x = 4 \sin \Theta$ show that $\cos \Theta = \frac{\sqrt{16-x^2}}{4}$ (5)

ii. By using the substitution $x = 4 \sin \Theta$

$$\text{show that } \int \frac{x^2 dx}{\sqrt{16-x^2}} = 8 \sin^{-1} \left(\frac{x}{4} \right) - \frac{x \sqrt{16-x^2}}{2} + c$$

- (c) The chord of contact of the tangents to the parabola $x^2 = 4ay$ from an external point $P(x_1, y_1)$ cuts the directrix at Q. Prove that PQ subtends a right angle at the focus of the parabola. (4)

Question 7 (12 marks)

- (a) Prove, using the Principle of Mathematical Induction, that
 $1 + 2 + 4 + \dots + 2^{n-1} = 2^n - 1$
for all positive integers $n \geq 1$ (3)

- (b) An projectile at the highest point of its trajectory has a velocity
8 metres per second and its position is 8 metres above the ground.
Find
i. the angle of projection (to nearest degree)
ii. the initial velocity (correct to 1 decimal place)
(take $g = 9.8 \text{ ms}^{-2}$) (5)

- (c)
i. Restrict the domain of $y = x^2 - 4x$ so that it will have an
inverse function of the largest possible domain, and will
include the point $x = 3$.
ii. Determine the equation of the inverse, writing y as the subject.
iii. Write down the co-ordinates of any points shared by the
original curve and its inverse. (4)

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1, \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x, \quad x > 0$

$$(a) \lim_{x \rightarrow 0} \frac{\sin 2x}{x}$$

$$= 2 \lim_{x \rightarrow 0} \frac{\sin 2x}{2x}$$

$$= 2$$

$$(c) A = 1, 3 = x_1, y_1$$

$$B = -2, 6 = x_2, y_2$$

$$\text{Divide in ratio} = -2:1$$

$$= k_1 : k_2$$

$$x = \frac{x_1 k_2 + x_2 k_1}{k_1 + k_2}$$

$$= \frac{(1 \times 1) + (-2 \times -2)}{-2 + 1}$$

$$= \frac{1 + 4}{-1}$$

$$= -5$$

$$y = \frac{y_1 k_2 + y_2 k_1}{k_1 + k_2}$$

$$= \frac{(3 \times 1) + (6 \times -2)}{-2 + 1}$$

$$= \frac{3 - 12}{-1}$$

$$= 9$$

$$\therefore (x, y) = (-5, 9)$$

$$(b) \int_0^{\pi} \sin^2 x \, dx$$

$$= \frac{1}{2} \int_0^{\pi} 1 - \cos 2x \, dx$$

$$= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi}$$

$$= \frac{1}{2} \left[(\pi - \frac{1}{2} \sin 2\pi) - (0) \right]$$

$$= \frac{\pi}{2}$$

$$ii) \int_{-1}^0 x \sqrt{1+x} \, dx$$

$$\text{Let } u = 1+x$$

$$\frac{du}{dx} = 1$$

$$\therefore du = 1 \, dx$$

$$x = -1, u = 0$$

$$x = 0, u = 1$$

$$\therefore \int_{-1}^0 x \sqrt{1+x} \, dx$$

$$= \int_0^1 (u-1) u^{\frac{1}{2}} \, du$$

$$= \int_0^1 u^{\frac{3}{2}} - u^{\frac{1}{2}} \, du$$

$$= \left[\frac{2u^{\frac{5}{2}}}{5} - \frac{2u^{\frac{3}{2}}}{3} \right]_0^1$$

$$= \frac{2}{5} - \frac{2}{3}$$

$$= -\frac{4}{15}$$

$$11 (d) a \cos \theta + b \sin \theta = A \cos(\theta - \alpha)$$

$$= A \cos \theta \cos \alpha + A \sin \theta \sin \alpha$$

$$a = A \cos \alpha, \quad b = A \sin \alpha$$

$$a^2 + b^2 = A^2 (\cos^2 \alpha + \sin^2 \alpha)$$

$$\therefore A = \sqrt{a^2 + b^2}$$

$$\frac{A \sin \alpha}{A \cos \alpha} = \frac{b}{a}$$

$$\therefore \tan \alpha = \frac{b}{a}$$

$$\text{Now solve } 3 \cos \theta + 4 \sin \theta = 2$$

$$A = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

$$\tan \alpha = \frac{4}{3} \quad \therefore \alpha = 0.9273$$

$$A \cos(\theta - \alpha) = 2$$

$$5 \cos(\theta - 0.9273) = 2$$

$$\cos(\theta - 0.9273) = \frac{2}{5}$$

$$\theta - 0.9273 = 1.1071 \quad \text{or} \quad 5.1239$$

$$\therefore \theta = 2.09 \quad \text{or} \quad 6.05 \quad (2.15)$$

$$\frac{1}{x} = 2 \sin x$$

$$x - 2 \sin x = 0$$

$$\text{Let } f(x) = x - 2 \sin x$$

$$f'(x) = 1 - 2 \cos x$$

$$\text{Let } a = 1.7 \text{ b.c. the}$$

first approx

$$a_1 = a - \frac{f(a)}{f'(a)}$$

$$= 1.7 - \frac{f(1.7)}{f'(1.7)}$$

$$= 1.9$$

$\therefore$ Second approximation to

the root is $x = 1.9$

$$\therefore -1 < x < 1 \quad \text{ans.}$$

$$62 (c) 4x - 3y + 1 = 0$$

$$4x + 1 = 3y$$

$$\therefore y = \frac{4}{3}x + \frac{1}{3} \quad (m_1 = \frac{4}{3})$$

$$x + 4y + 1 = 0$$

$$4y = -x - 1$$

$$\therefore y = -\frac{1}{4}x - \frac{1}{4} \quad (m_2 = -\frac{1}{4})$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{4}{3} - (-\frac{1}{4})}{1 + \frac{4}{3} \times (-\frac{1}{4})} \right|$$

$$= \left| \frac{2 \frac{1}{4}}{\frac{1}{3}} \right|$$

$$\theta = 67^\circ \quad \text{ans.}$$

$$(d) P(x) = x^3 + ax^2 + bx - 18$$

$(x+2)$ is a factor

$$\therefore P(-2) = -8 + 4a - 2b - 18 = 0$$

$$4a - 2b = 26$$

$$2a - b = 13$$

(1)

Remainder = -24 when $P(x)$ is divided by $(x-1)$

$$\therefore P(1) = 1 + a + b - 18 = -24$$

$$a + b = -7$$

(2)

$$\text{Add (1) + (2)}$$

$$3a = 6$$

$$\therefore a = 2$$

ans.

(a) $y = 2 \sin^{-1} x$

$\frac{dy}{dx} = \frac{2}{\sqrt{1-x^2}}$

Let $x = \frac{1}{\sqrt{2}}$, $\frac{dy}{dx} = \frac{2}{\sqrt{1-\frac{1}{2}}} = \frac{2}{\frac{1}{\sqrt{2}}} = 2\sqrt{2}$

Eqn of tangent is

$y - y_1 = m(x - x_1)$

$y - \frac{\pi}{2} = 2\sqrt{2}(x - \frac{1}{\sqrt{2}})$

$y - \frac{\pi}{2} = 2\sqrt{2}x - 2$

$\therefore y = 2\sqrt{2}x + \frac{\pi}{2} - 2$

Put $x = \frac{1}{\sqrt{2}}$

$y = 2 \sin^{-1} \frac{1}{\sqrt{2}} = 2 \cdot \frac{\pi}{4} = \frac{\pi}{2}$

$(x-9)(x-1)$

$x = 9$ or -1

It can't get to $x = -1$ if it does not pass through $x = 0$

$\therefore x = 9$ m

Q3 (a) $\frac{dV}{dt} = 200$

$V = \frac{4}{3} \pi r^3$

$\frac{dV}{dr} = 4\pi r^2$

$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$

$200 = 4\pi r^2 \cdot \frac{dr}{dt}$

$\therefore \frac{dr}{dt} = \frac{200}{4\pi r^2} = \frac{50}{\pi r^2}$

ii) $S = 4\pi r^2$

$\frac{dS}{dr} = 8\pi r$

$\frac{dS}{dt} = \frac{dS}{dr} \cdot \frac{dr}{dt}$

$= 8\pi r \cdot \frac{50}{\pi r^2}$

$= \frac{400}{r}$

put $r = 50$

$\therefore \frac{dS}{dt} = \frac{400}{50} = 8 \text{ cm}^2/\text{s}$

$\angle PNQ = \angle SNR$ (vert opp ang)

$\angle SMR = \angle PMQ$ (PM and SM are of same line)

$\therefore \angle PNQ + \angle PMQ = \theta$

Q4 (a) Prove $\frac{2 \tan \theta}{1 + \tan^2 \theta} = \sin 2\theta$

L.H.S. $= \frac{2 \tan \theta}{\sec^2 \theta}$

$= 2 \frac{\sin \theta}{\cos \theta} \cdot \cos^2 \theta$

$= 2 \sin \theta \cos \theta$

$= \sin 2\theta$

$= \text{R.H.S.}$

(b) $f(x) = 2 \cos^{-1} \frac{x}{3}$

$\therefore f(0) = 2 \cos^{-1} 0 = 2 \cdot \frac{\pi}{2} = \pi$

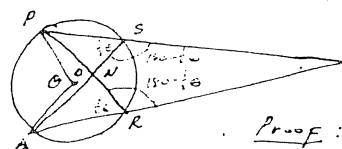
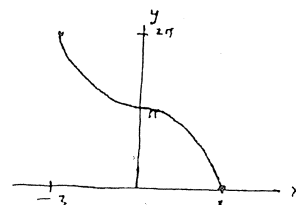
ii) $-1 \leq \frac{x}{3} \leq 1$

$\therefore -3 \leq x \leq 3$ is domain of $f(x)$

$0 \leq \cos^{-1} \frac{x}{3} \leq \pi$

$\therefore 0 \leq 2 \cos^{-1} \frac{x}{3} \leq 2\pi$ is Range of $f(x)$

iii)



Qim: Prove $\angle PRM = (180 - \frac{1}{2}\theta)$

Proof: $\angle PRQ = \frac{1}{2}\theta$ (Lat centre is $2 \times$ Lat at circum)
 $\therefore \angle PRM = 180 - \frac{1}{2}\theta$ (adj supp $\angle$ s)

Qim: Prove $\angle PNA + \angle PMQ = \theta$

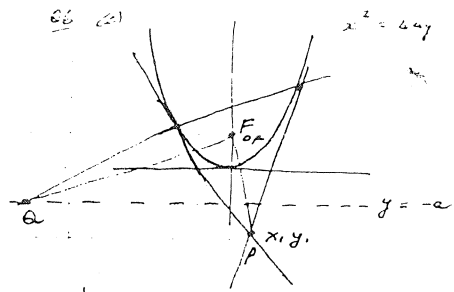
Proof: $\angle PSB = \frac{1}{2}\theta$ (Lat Centre is $2 \times$ Lat at circum)

$\therefore \angle GSM = 180 - \frac{1}{2}\theta$ (adj supp $\angle$ s)

$\angle SNR + \angle SMR = 360 - (180 - \frac{1}{2}\theta) - (180 - \frac{1}{2}\theta)$ (L sum of quad)
 $= \theta$

Q6

$x = 4 \sin \theta$
 $\frac{dx}{d\theta} = 4 \cos \theta$
 $\therefore dx = 4 \cos \theta d\theta$
 $\therefore x^2 = 16 \sin^2 \theta$



$$\int \frac{x^2 dx}{\sqrt{16-x^2}} = \int \frac{16 \sin^2 \theta \cdot 4 \cos \theta d\theta}{\sqrt{16-16 \sin^2 \theta}}$$

$$= \int \frac{16 \sin^2 \theta \cdot 4 \cos \theta d\theta}{4 \cos \theta}$$

$$= 16 \int \sin^2 \theta d\theta$$

$$= 16 \cdot \frac{1}{2} \int (1 - \cos 2\theta) d\theta$$

$$= 8 \left[\theta - \frac{1}{2} \sin 2\theta \right] + C$$

$$= 8 \left[\theta - \frac{1}{2} \cdot 2 \sin \theta \cos \theta \right] + C$$

$$8\theta - 8 \sin \theta \cos \theta + C$$

$$8 \sin^{-1} \frac{x}{4} - 8 \frac{x}{4} \cdot \frac{\sqrt{16-x^2}}{4} + C$$

$$8 \sin^{-1} \frac{x}{4} - \frac{1}{2} x \sqrt{16-x^2} + C$$

$\cos 2\theta = 1 - 2 \sin^2 \theta$
 $2 \sin^2 \theta = 1 - \cos 2\theta$
 $\sin^2 \theta = \frac{1}{2} (1 - \cos 2\theta)$

$\sin \theta = \frac{x}{4}$
 $\theta = \sin^{-1} \left(\frac{x}{4} \right)$

$\cos \theta = \frac{\sqrt{16-x^2}}{4}$
 From part (i)

Chord of contact from $P(x_1, y_1)$ is

$xx_1 = 2a(y+y_1)$

This cuts directrix ($y = -a$) at Q

$xx_1 = 2a(y_1 - a)$

$x = \frac{2a(y_1 - a)}{x_1}$

$\therefore \text{Point } Q = \left[\frac{2ay_1 - 2a^2}{x_1}, -a \right]$

Grad PF = $\frac{(y_1 - a)}{x_1}$ (m_1)

Grad QF = $\frac{-a - a}{\left(\frac{2ay_1 - 2a^2}{x_1} \right)} = \frac{-2ax_1}{2ay_1 - 2a^2}$ (m_2)

$m_1 \times m_2 = \frac{(y_1 - a)}{x_1} \cdot \frac{-2ax_1}{2a(y_1 - a)} = -1$

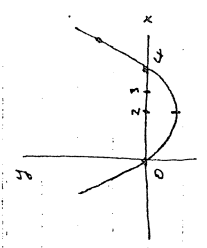
$\therefore PF \perp QF$

Thus PQ subtends a right angle at the focus

Q7 a Prove $1+2+4+\dots+2^{n-1} = 2^n - 1$ for $n \geq 1$

Step 1: Prove true for $n=1$
 L.H.S = 1 R.H.S = $2^1 - 1 = 1$
 $\therefore$ True for $n=1$
 Step 2: Assume true for $n=k$
 $\therefore 1+2+4+\dots+2^{k-1} = 2^k - 1$
 Step 3: Prove true for $n=k+1$ i.e. Prove $1+2+4+\dots+2^k = 2^{k+1} - 1$
 L.H.S = $1+2+4+\dots+2^{k-1} + 2^k$
 $= (2^k - 1) + 2^k$
 $= 2^k(1+1) - 1$
 $= 2^{k+1} - 1$
 R.H.S = $2^{k+1} - 1$
 $\therefore$ True for $n=k+1$

Step 4: If true for $n=1$, then true for $n=2$ etc.
 Hence by Math. Induction $1+2+4+\dots+2^{n-1} = 2^n - 1$ for all $n \geq 1$



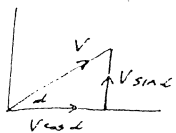
Q7(c) $y = x^2 - 4x$
 $= x(x-4)$
 i. Inverse Domain = $x \geq 2$
 ii. Complete the square
 $4+x = y^2 - 4y + 4$
 $\therefore (y-2)^2 = x+4$
 $y-2 = \pm \sqrt{x+4}$
 $\therefore y = 2 \pm \sqrt{x+4}$

Test $x=5$, $y = 2 + \sqrt{9} = 5$
 $\therefore$ Inverse function is $y = 2 + \sqrt{x+4}$

iii. To find the common point.

Solve $y = x$
 $y = x^2 - 4x$
 $x^2 - 4x = x$
 $\therefore x^2 - 5x = 0$
 $x(x-5) = 0$
 $\therefore x = 5, y = 5$ Ans (5,5)

7d)



Vertical motion. Take up ↑ positive

Data: $t=0, y=0, \dot{y} = V \sin \alpha$

$$\ddot{y} = -g$$

$$\dot{y} = -gt + C$$

$$V \sin \alpha = 0 + C$$

$$\therefore \dot{y} = -gt + V \sin \alpha \quad (*)$$

$$y = -\frac{gt^2}{2} + Vt \sin \alpha + C$$

$$0 = 0 + 0 + C$$

$$\therefore y = -\frac{gt^2}{2} + Vt \sin \alpha \quad (*)$$

Horizontal motion Data: $t=0, x=0, \dot{x} = V \cos \alpha$

$$\ddot{x} = 0$$

$$\dot{x} = 0 + C$$

$$V \cos \alpha = C$$

$$\therefore \dot{x} = V \cos \alpha \quad (*)$$

$$x = Vt \cos \alpha + C$$

$$0 = 0 + C$$

$$\therefore x = Vt \cos \alpha \quad (*)$$

67b i

At highest point $\dot{y} = 0$

$$0 = -gt + V \sin \alpha$$

$$gt = V \sin \alpha$$

$$t = \frac{V \sin \alpha}{g}$$

At highest point $y = 8$

$$8 = -\frac{gt^2}{2} + Vt \sin \alpha$$

$$= -\frac{g}{2} \frac{V^2 \sin^2 \alpha}{g^2} + V \sin \alpha \frac{V \sin \alpha}{g}$$

$$= -\frac{V^2 \sin^2 \alpha}{2g} + \frac{V^2 \sin^2 \alpha}{g}$$

$$\therefore 8 = \frac{V^2 \sin^2 \alpha}{2g} \quad (1)$$

Also at highest point $\dot{x} = 8$

$$8 = V \cos \alpha$$

$$\therefore V = \frac{8}{\cos \alpha}$$

Sub into (1) at

$$8 = \frac{64}{\cos^2 \alpha} \cdot \frac{\sin^2 \alpha}{2g}$$

$$\frac{16 \times 9.8}{64} = \frac{\sin^2 \alpha}{\cos^2 \alpha} = \tan^2 \alpha$$

$$2.45 = \tan^2 \alpha$$

$$\therefore \tan \alpha = \sqrt{2.45} = 1.5652$$

$$\alpha = 57^\circ 26' \approx 57^\circ = \text{angle of projection}$$

$$\text{So } V = \frac{8}{\cos \alpha} = 14.7 \text{ m/s}$$

= initial velocity